# DATA QUALITY ASSESSMENT & MONITORING FRAMEWORK

### A Data-Driven Approach to Identifying and Managing Operational Data Issues

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**Technology:** Python | Pandas | Google Colab

**Datasets Analyzed:** 18

**Total Records:** 639,328

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## 1. Project Background

Modern organizations depend on data from multiple operational systems such as customer management, payments, calls, agents, campaigns, and communication platforms.

When data comes from different sources, problems such as **missing information, duplicate records, inconsistent identities, and conflicting transaction references** can occur.

These issues can affect reporting, customer communication, payment reconciliation, and business decisions.

This project was developed to systematically **profile, validate, identify, and analyze data-quality issues** across multiple operational datasets.

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## 2. Problem Statement

The main problem addressed by this project is:

**How can data-quality issues across multiple operational datasets be systematically identified, measured, and prioritized before they impact business operations?**

The project focuses on detecting:

- Incomplete data
- Duplicate records
- Duplicate identifiers
- Borrower identity conflicts
- Agent identity conflicts
- Payment-reference conflicts
- Invalid timestamps

- Campaign-targeting patterns

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## 3. Project Objective

The objective is not simply to find errors.

The objective is to build a **structured data-quality assessment process** that can:

**Detect → Measure → Understand → Prioritize → Recommend Improvements**

This helps transform raw operational data into information that can be trusted for analysis and decision-making.

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## 4. Data Environment

The analysis covered **18 operational datasets** and **639,328 records**.

The datasets represent different operational areas including:

- Borrowers
- Accounts
- Payments
- Calls
- Agents
- Agent Sessions
- Field Visits
- Complaints
- Campaigns
- SMS Events
- WhatsApp Events
- Daily Targeting
- Promises to Pay

### Why multiple datasets matter

Data-quality problems are not always visible within a single table.

For example:

A borrower may have the same borrower_id but different identity information in different records.

Similarly:

A payment reference may appear multiple times and create uncertainty during transaction reconciliation.

Therefore, the project performs both **individual dataset checks and cross-record consistency analysis**.

## 5. Methodology

The project follows a structured six-stage approach:

**STEP 1 — Data Profiling**

Understand dataset size, columns, identifiers, and overall structure.

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**STEP 2 — Completeness Analysis**

Identify missing values in important fields.

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**STEP 3 — Uniqueness Analysis**

Detect duplicate records and repeated identifiers.

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**STEP 4 — Consistency Analysis**

Check borrower, agent, and payment identity consistency.

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**STEP 5 — Validation**

Check timestamps and other logical conditions.

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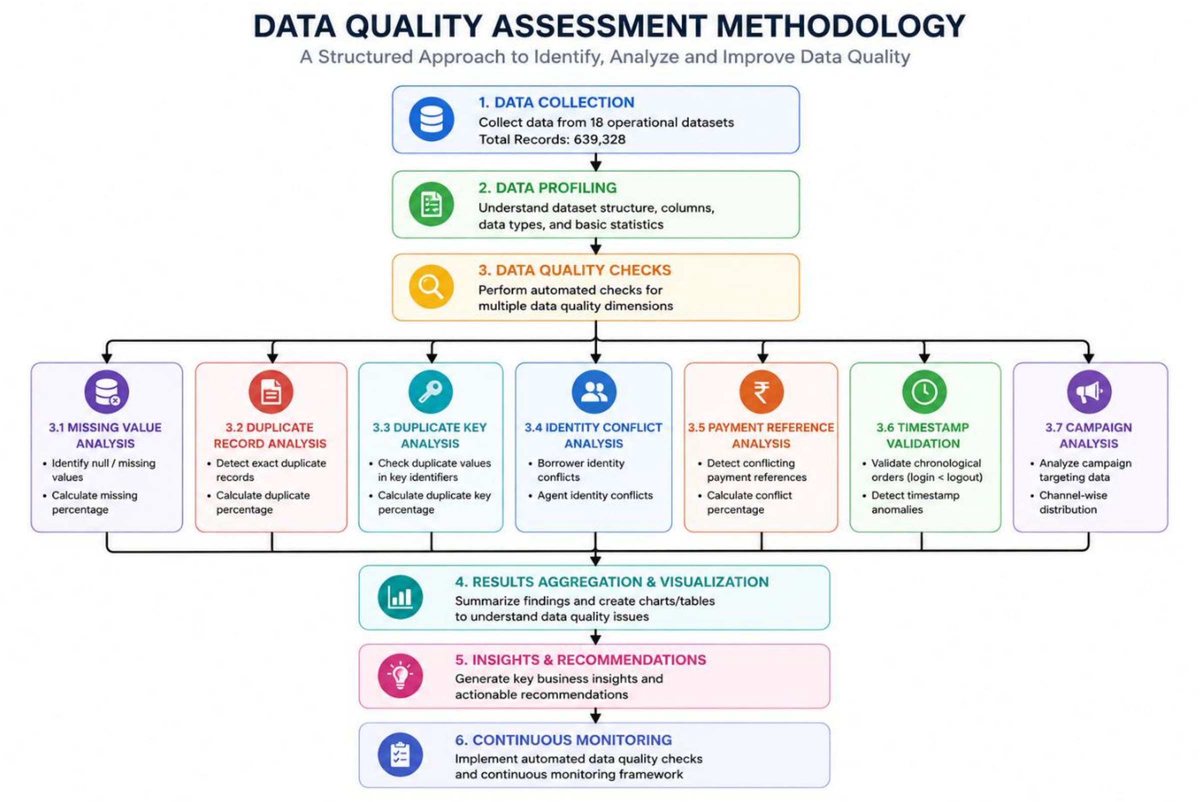
**STEP 6 — Business Interpretation**

Convert technical findings into actionable recommendations.

### Figure 1: Data Quality Assessment Methodology

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## 6. Data Quality Dimensions

The project evaluates data using five major dimensions:

| Dimension | What Was Checked |
|---|---|
| **Completeness** | Missing values |
| **Uniqueness** | Duplicate records and keys |
| **Consistency** | Identity and payment conflicts |
| **Validity** | Timestamp relationships |
| **Usability** | Campaign and operational analysis |

This provides a more structured assessment than checking only for missing values.

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## 7. Completeness Analysis

Missing-value analysis identified incomplete information in several important fields.

### Major observations

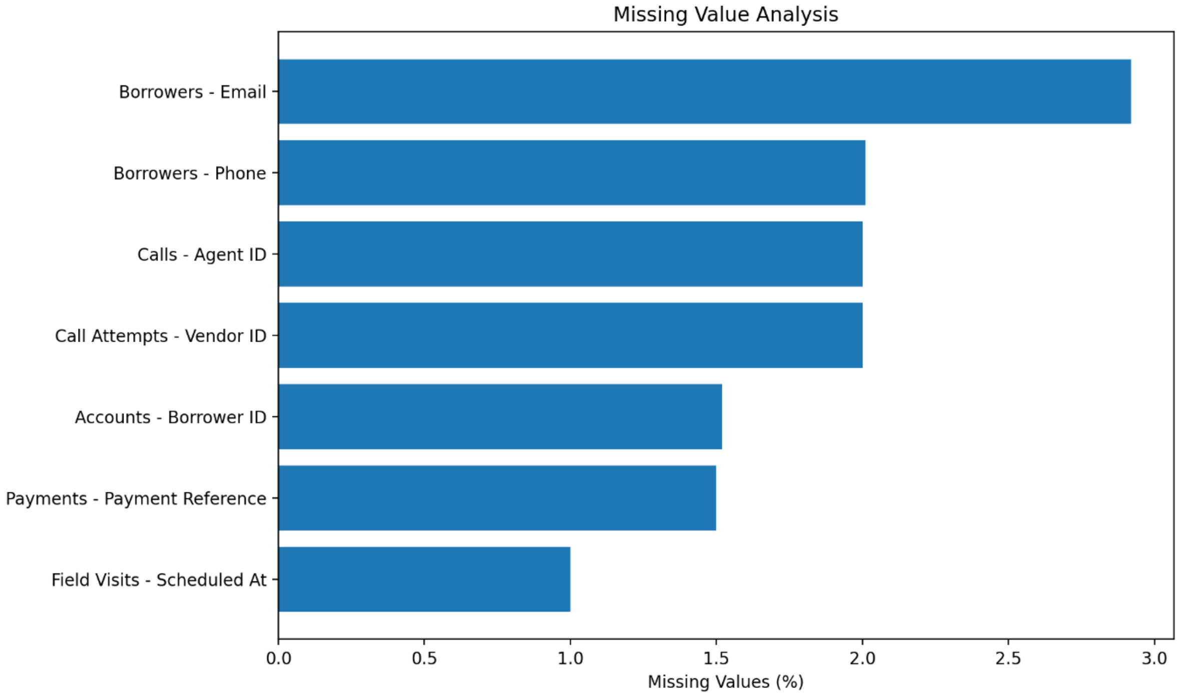
- Borrower email missing: **2.92%**
- Borrower phone missing: **2.01%**
- Agent ID missing in calls: **2.00%**
- Vendor ID missing in call attempts: **2.00%**
- Borrower ID missing in accounts: **1.52%**
- Payment reference missing: **1.50%**
- Scheduled timestamp missing in field visits: **1.00%**

### Interpretation

Although the percentages appear relatively small, these fields are operationally important.

For example, missing borrower contact information can affect customer communication, while missing identifiers can prevent accurate linking between datasets.

Figure 2: Missing Value Analysis



## 8. Uniqueness Analysis

Duplicate analysis was performed at two levels:

### Level 1 — Exact Duplicate Records

| Dataset | Duplicate % |
|---|---|
| **Borrowers** | 1.96% |
| **Payments** | 1.91% |
| **Calls** | 1.39% |
| **WhatsApp Events** | 0.99% |

### Level 2 — Duplicate Identifiers

| Identifier | Duplicate % |
|---|---|
| **Borrower ID** | **64.00%** |
| **Payment Reference** | **18.35%** |
| **Payment ID** | 1.96% |
| **Call ID** | 1.48% |
| **WhatsApp Event ID** | 0.99% |

### Important Interpretation

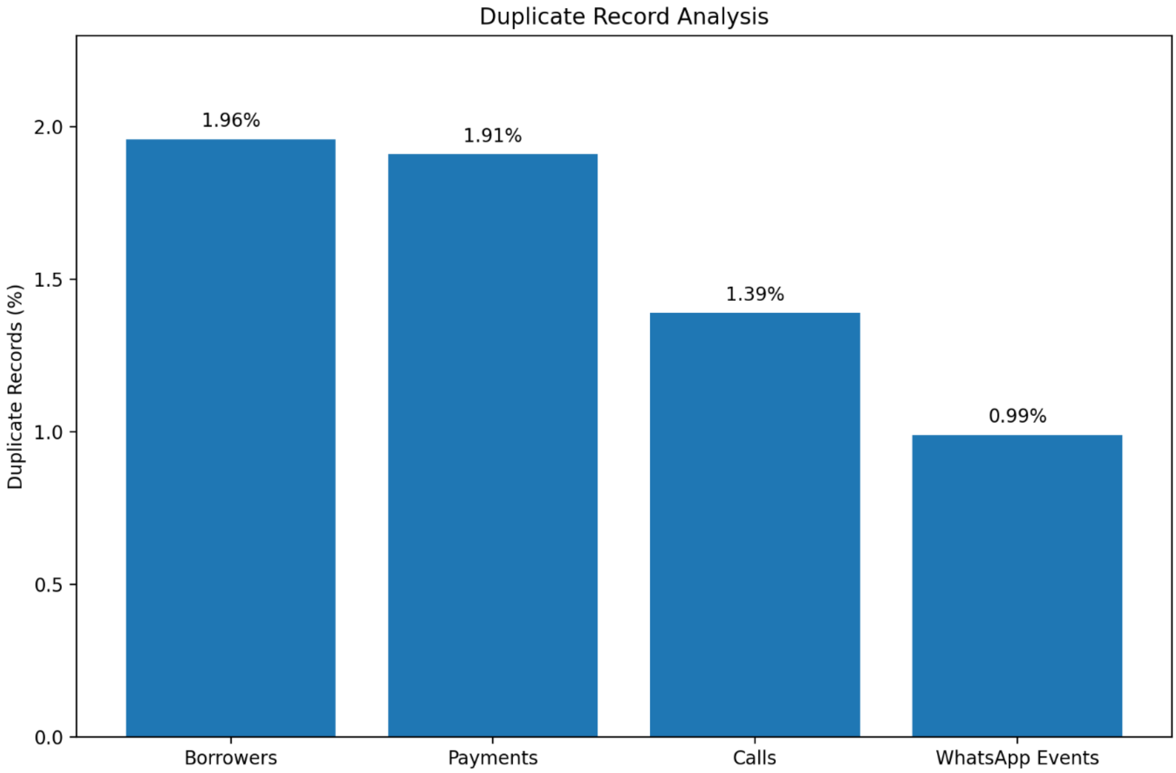
A repeated identifier is **not automatically an error**.

For example, the same borrower can legitimately appear in multiple transactions or activities.

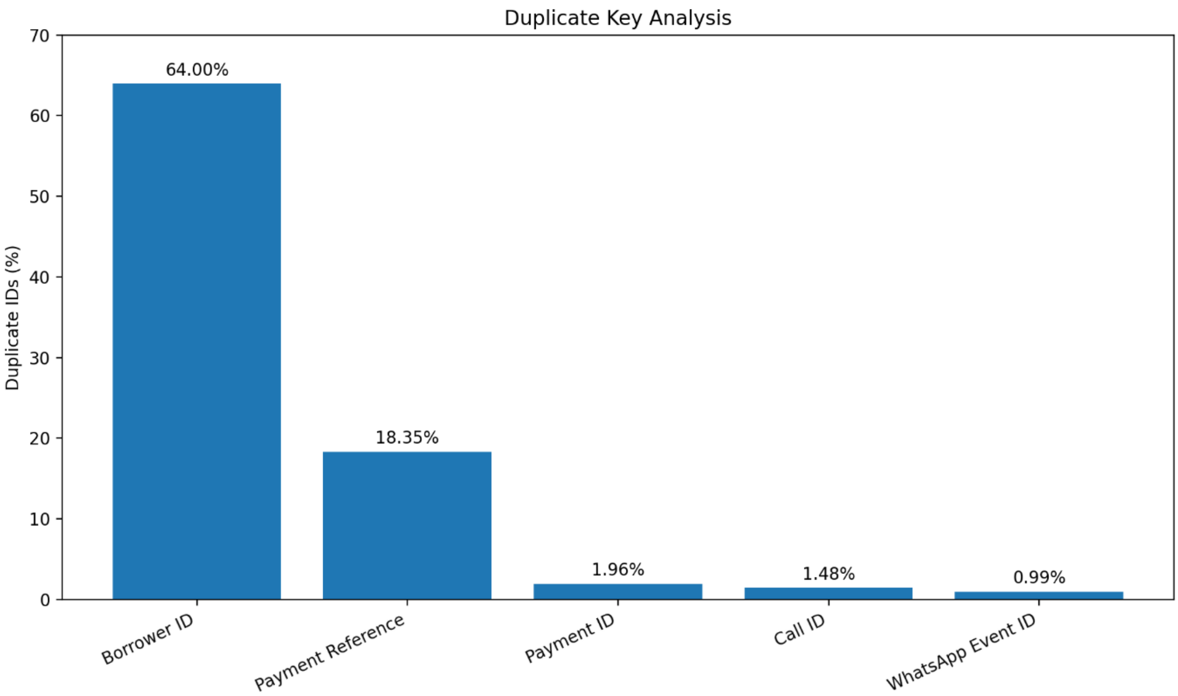
Therefore, the project goes one step further and checks whether repeated identifiers are associated with **conflicting identity information**.

This is an important distinction in data-quality analysis.

### Figure 3: Duplicate Record Analysis




### Figure 4: Duplicate Key Analysis



## 9. Identity Resolution Analysis

Identity consistency was one of the most important areas of the project.

### Borrower Identity

The analysis identified:

8,518 conflicting borrower IDs out of 11,015 unique borrower IDs

Conflict Rate = 77.33%

This indicates substantial inconsistency among identity attributes associated with borrower IDs.

### Potential Impact

- Incorrect customer profiles
- Duplicate customer representation
- Incorrect customer communication
- Unreliable borrower-level analytics
- Difficulty establishing a single customer view

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### Agent Identity

The analysis reported:

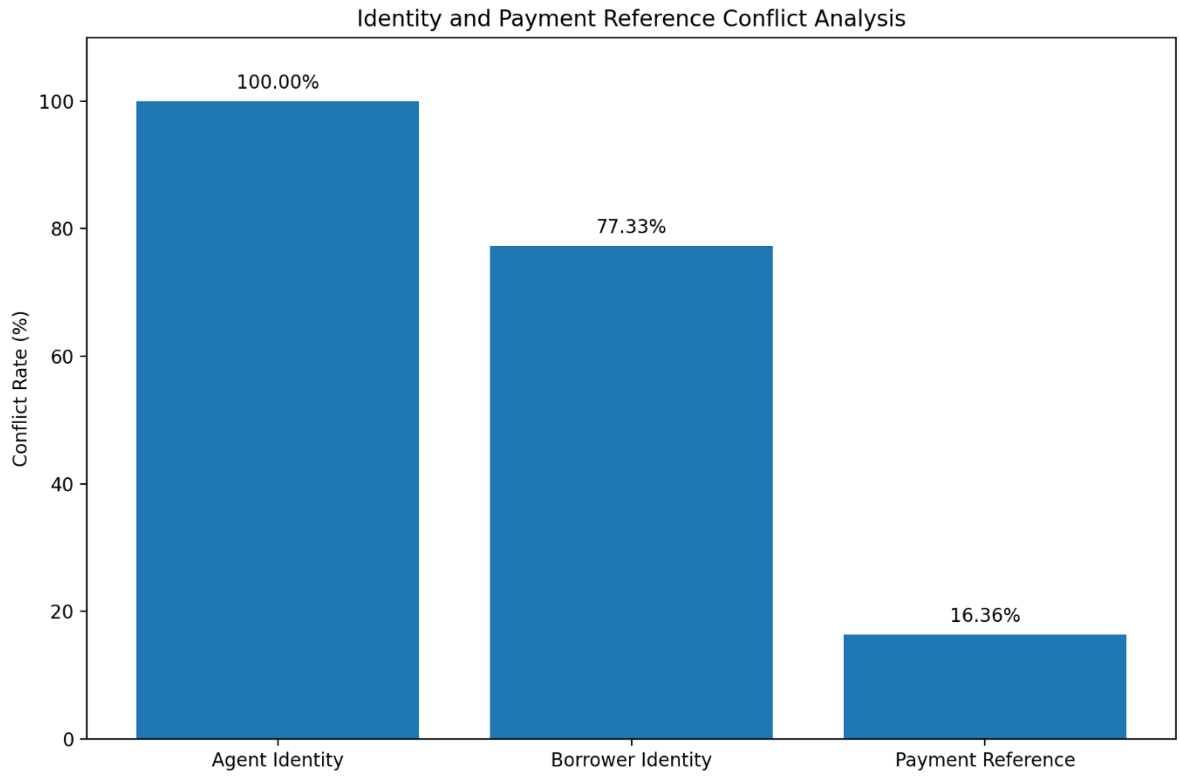
1,000 agent IDs with identity conflicts

Conflict Rate = 100%

This is a critical finding because agent identity is important for:

- Performance measurement
- Call attribution
- Productivity analysis
- Accountability
- Operational reporting

**Figure 5: Identity and Payment Reference Conflict Analysis**




## 10. Payment Integrity Analysis

Payment data was analyzed separately because transaction accuracy is critical.

**Findings**

**Unique payment references:** 20,821

**Conflicting payment references:** 3,407

**Conflict rate:** 16.36%

Additionally, exact duplicate payment records represented:

**486 out of 25,500 payment rows**

**Duplicate rate:** 1.91%

**Business Risk**

Payment-reference inconsistencies can affect:

- Payment reconciliation
- Transaction tracking
- Financial reporting

- Duplicate-payment detection
- Auditing

Therefore, payment references should be governed using clear uniqueness and validation rules.

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## 11. Timestamp Validation

The project also performed a logical timestamp validation.

**Validation Rule**

Logout time should not occur before login time.

**Result**

**Invalid agent sessions detected: 0**

This indicates that the specific chronological validation rule passed successfully.

This is an example of a business-rule-based data-quality check, rather than a simple statistical check.

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## 12. Campaign Targeting Analysis

The project also examined the operational targeting data.

**Dataset Statistics**

**Targeting records**: 45,000

**Unique targeted accounts:** 23,344

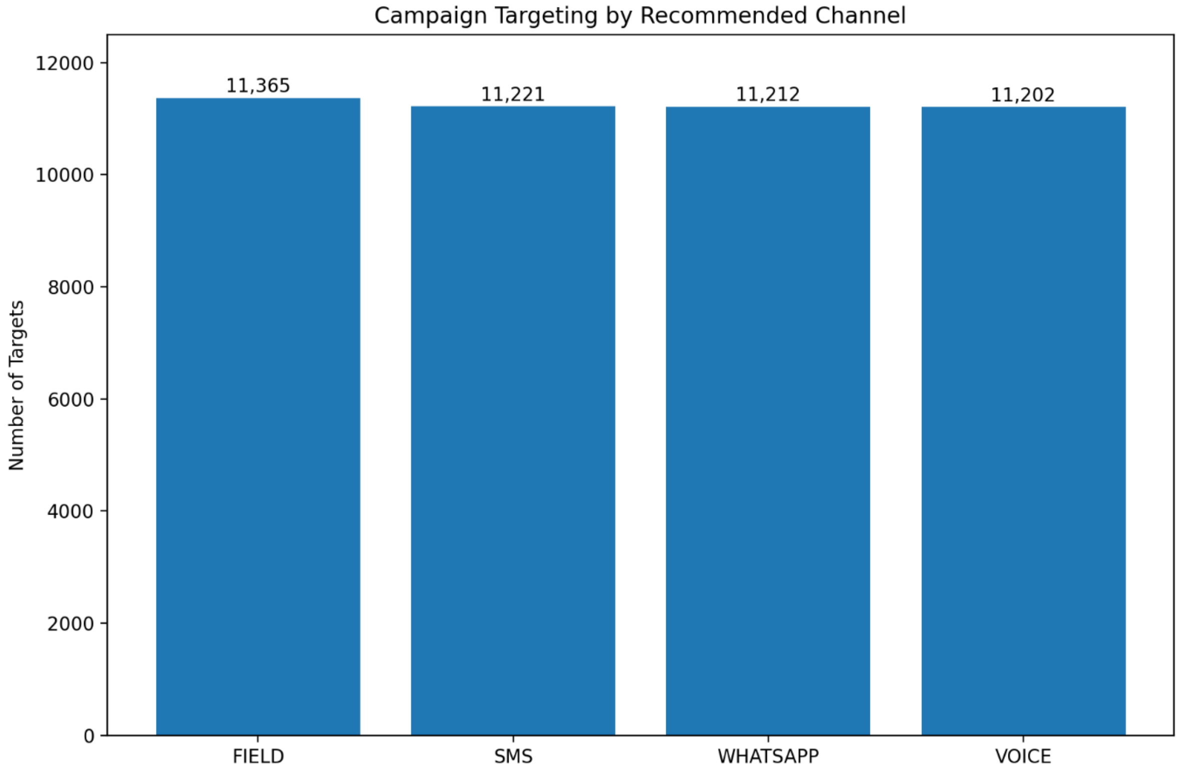
**Campaigns: 120**

**Channel Distribution**

| Channel | Targets |
|---|---|
| **FIELD** | **11,365** |
| **SMS** | **11,221** |
| **WHATSAPP** | **11,212** |
| **VOICE** | **11,202** |

The targeting distribution is relatively balanced across the four channels.

### Figure 6: Campaign Targeting by Communication Channel




## 13. Data Quality Risk Assessment

The findings can be prioritized based on their potential business impact.

### High Priority

**Agent identity conflicts** — 100%

**Borrower identity conflicts** — 77.33%

**Borrower ID repetition** — 64.00%

**Payment-reference conflicts** — 16.36%

These issues can directly affect identity management, reporting, and transaction reliability.

### Medium Priority

**Payment duplicate records** — 1.91%

**Borrower duplicate records** — 1.96%

**Missing critical fields** — up to 2.92%

### Validated Area

**Logout-before-login violations** — 0

## 14. Business Impact

The identified issues can have consequences beyond the database itself.

| Data Issue | Potential Business Impact |
|---|---|
| **Identity conflicts** | Incorrect customer/agent attribution |
| **Duplicate records** | Inflated reporting |
| **Payment conflicts** | Reconciliation problems |
| **Missing identifiers** | Broken relationships between datasets |
| **Missing contact details** | Communication failures |
| **Invalid timestamps** | Incorrect activity timelines |

Therefore, improving data quality can directly improve **operational reliability and decision-making**.

## 15. Recommended Solution

A sustainable solution should not depend only on manually cleaning the current data.

The recommended approach is to introduce **automated Data Quality Controls.**

### Proposed Framework

Data Ingestion

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Automated Validation

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Quality Score Calculation

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Issue Classification

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Data Remediation

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Monitoring Dashboard

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Continuous Improvement

### Recommended automated checks

- Required-field validation
- Duplicate detection
- Identifier uniqueness checks
- Identity consistency checks
- Payment-reference validation
- Timestamp validation
- Referential-integrity checks
- Data-quality threshold alerts

## 16. Key Insights

The project produced five major insights:

- **1.Identity consistency is the most critical issue.**
  The borrower and agent datasets show substantial identity conflicts.
- **2.Duplicate identifiers require contextual analysis.**
  Repeated IDs should not automatically be deleted because some may represent legitimate transactions or activities.
- **3.Payment data requires stronger controls.**
  Conflicting payment references can create reconciliation and reporting risks.
- **4.Missing values are concentrated in operationally important fields.**
  Even relatively small percentages can have significant business impact when the affected fields are identifiers or contact information.
- **5.Automated monitoring is necessary.**
  Data quality should be treated as an ongoing process rather than a one-time cleaning activity.

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## 17. Key Findings

Use these findings from your actual results:

- **77.33% of unique borrower IDs have identity conflicts**, indicating major borrower identity inconsistency.
- **100% of agent IDs show identity conflicts**, requiring strong agent identity validation.

- **16.36% of payment references have conflicts**, creating a risk of incorrect payment mapping.
- **Borrowers have 600 duplicate records (1.96%)**.
- **Payments have 486 duplicate records (1.91%)**.
- **Calls have 1,271 duplicate records (1.39%)**.
- **WhatsApp events have 600 duplicate records (0.99%)**.
- The highest missing-value issue is **borrower email: 895 records (2.92%)**.
- **Call attempts have 2,400 missing vendor IDs (2.00%)**.
- No timestamp ordering violation was detected in the agent-session analysis.
- Campaign targeting contains **45,000 records across 120 campaigns**.

## 18. Recommendations

1. Implement unique borrower and agent identity validation.
2. Standardize and validate payment references before reporting.
3. Detect and remove duplicate transaction records.
4. Improve collection of borrower phone numbers and email addresses.
5. Validate mandatory foreign-key fields such as agent_id, vendor_id, and borrower_id.
6. Add automated data-quality monitoring for duplicates and identity conflicts.
7. Use campaign/channel performance analysis to improve targeting decisions.

## Tools & Technologies

| Category | Technology |
|---|---|
| **Programming** | Python |
| **Data Analysis** | Pandas |
| **Numerical Analysis** | NumPy |
| **Visualization** | Matplotlib |
| **Development Environment** | Google Colab |
| **Approach** | Data Profiling & Data Quality Validation |

## Project Limitations

- Some duplicate identifiers may represent legitimate repeated transactions or activities and therefore require business validation before removal.
- The analysis identifies and measures data-quality issues, while complete data remediation would require integration with the original source systems.

- The assessment is based on the datasets provided for the project.

### Future Enhancements

- Develop an automated data-quality dashboard.
- Introduce real-time validation during data ingestion.
- Implement automated alerts for critical data-quality violations.
- Create a data-quality score for each dataset.
- Enable continuous monitoring of identity, duplicate, and payment-reference issues.

## 19. Technology Stack

Python
Pandas
Google Colab
Matplotlib
Data Profiling
Data Validation
Data Visualization

## 20. Project Limitations

- The analysis focuses on identifying and measuring data-quality issues in the available datasets.
- Some duplicate identifiers may represent legitimate repeated transactions or activities and therefore require business validation before removal.
- The project identifies quality issues and recommends controls; complete remediation would require integration with the source systems.

## 21. Future Enhancements

- Develop an automated data-quality dashboard.
- Introduce real-time validation during data ingestion.
- Implement automated alerts for critical quality violations.
- Create a centralized data-quality score for each dataset.
- Add continuous monitoring of identity and payment conflicts.

## 22. Final Project Takeaway

This project demonstrates that data quality is not only about removing missing or duplicate records. It requires systematic validation of completeness, uniqueness, consistency, identity, transaction integrity, and business rules. The proposed framework **can help organizations**

**detect critical data issues early and improve the reliability of operational decision-making.**

## Conclusion

The Data Quality Assessment successfully evaluated 18 operational datasets containing 639,328 records. The analysis identified key issues related to missing values, duplicate records, duplicate identifiers, identity conflicts, payment-reference inconsistencies, and timestamp quality.

The most significant findings were the high borrower and agent identity conflict rates and payment-reference inconsistencies. These issues can affect customer identification, operational reporting, payment reconciliation, and business decision-making.

The project demonstrates that systematic data profiling, validation, visualization, and continuous monitoring can improve data reliability. Implementing the recommended data-quality controls can help organizations identify critical issues early and maintain accurate and trustworthy operational data.